MILITARY SPECIFICATION

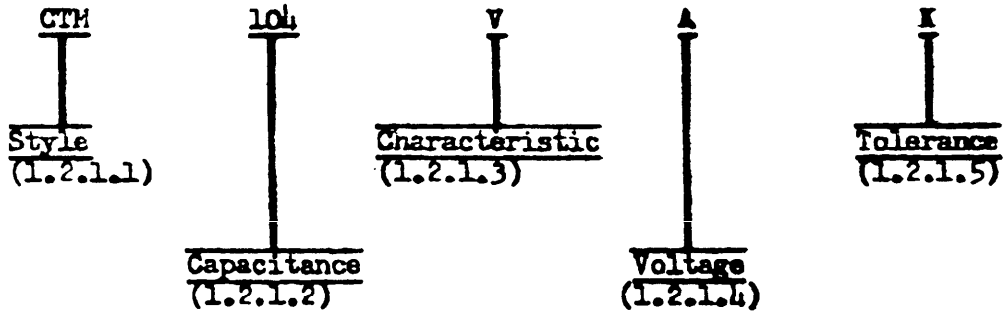
CAPACITORS, FIXED, PLASTIC DIELECTRIC, DC (NONMETALLIC CASES), GENERAL SPECIFICATION FOR

1. SCOPE

1.1 SCOPE.- This specification covers direct-current, plastic dielectric, fixed capacitors with nonmetallic cases. These capacitors are intended for operation in ambients from -55° to + 125°C.

1.2 CLASSIFICATION

1.2.1 TYPE DESIGNATION.- The type designation shall be in the following form, and as specified:



1.2.1.1 STYLE.- The style is identified by the three-letter symbol "CTM;" the three letters "CTM" identify dc, plastic dielectric, fixed capacitors enclosed in nonmetallic tubular cases.

1.2.1.2 CAPACITANCE.- The nominal capacitance value expressed in micromicrofarads is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow.

1.2.1.3 CHARACTERISTIC.- The characteristic is identified by a single letter in accordance with the applicable detail specification.

1.2.1.4 VOLTAGE.- The dc working voltage for continuous operation at 125°C is identified by a single letter in accordance with table I.

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TABLE I. - Voltage Rating

Symbol	D.C. Voltage Rating at 125°C
A	50
B	100
C	200
D	300

1.2.1.5 CAPACITANCE TOLERANCE.- The capacitance tolerance in percent is identified by a single letter in accordance with table II.

TABLE II - CAPACITANCE TOLERANCE

Symbol	Tolerance in Percent
J	± 5
K	± 10

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-B-665	Boxes, Paperboard, Metal-Stayed (Including stayed material)
PPP-B-676	Boxes, Set-Up, Paperboard
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof - for Packaging and Sealing
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced

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Military

MIL-P-116	Preservation, Methods of
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid
MIL-L-10547	Liners, Case, Waterproof

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 OTHER PUBLICATIONS.- The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Consolidated Classification Committee - Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 QUALIFICATION.- The capacitors furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 DETAIL SPECIFICATIONS FOR INDIVIDUAL CAPACITOR TYPES.- All requirements specified herein apply to all types of capacitors covered by applicable detail specifications. Detail requirements or exceptions applicable to individual types will be specified in the detail specifications. In the event of any conflict between requirements of this specification and the individual detail specification, the latter shall govern.

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3.3 MATERIALS.- When a definite material is not specified, a material shall be used that will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 SEALING COMPOUNDS.- Compounds used in the sealing of capacitors shall be chemically inactive with respect to the capacitor element and the case. The compound, either in the state of original application or as a result of having aged, shall have no adverse effect on the performance of the capacitor.

3.4 DESIGN AND CONSTRUCTION.- Capacitors shall be of the design, construction, and physical dimensions specified (see 3.2).

3.4.1 CASE.- Each capacitor shall be enclosed in a nonmetallic case which will protect the capacitor element from moisture and mechanical damage under all the test conditions specified herein. The use of exterior cardboard sleeves for insulating purposes shall not be permitted.

3.4.2 TERMINALS

3.4.2.1 LEADS.- Leads shall be of copper, copper-alloy, or copper-clad steel, axially located. Copper alloy and copper-clad leads shall provide a minimum of 30 percent of the conductivity of electrolytic copper.

3.4.2.2 TERMINAL INSULATOR.- The terminal insulator shall be thermo setting resin.

3.5 DIELECTRIC WITHSTANDING VOLTAGE AT ROOM TEMPERATURE.- Capacitors shall withstand the potentials specified in the applicable detailed specification without momentary or intermittent arcing nor any other indication of breakdown or arcing during the test. As a result of the test, there shall be no open or short-circuiting not permanent damage to the capacitor.

3.6 BAROMETRIC PRESSURE (FLASHOVER).- Capacitors shall withstand the potentials specified in 4.6.3 without momentary or intermittent arcing or any other indication of breakdown or arcing during the test. As a result of the test, there shall be no open or short-circuiting nor permanent damage to the capacitor.

3.7 INSULATION RESISTANCE

3.7.1 TERMINAL TO TERMINAL.- When measured, the insulation resistance shall not be less than the values shown in the applicable detail specification. The value of insulation resistance varies with

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temperature, and it is necessary to apply a correction factor if measurements are made at a capacitor temperature other than 25°C. Table III gives these correction factors for measurements made with the temperature range from 20° to 35°C. The required value of insulation resistance shall be multiplied by the correction factor to determine the new value required at the test temperature.

TABLE III.- Insulation Resistance Correction Factors

DEGREES C	CORRECTION FACTOR
20	1.4
21	1.3
22	1.2
23	1.1
24	1.0
25	1.0
26	0.94
27	0.87
28	0.82
29	0.76
30	0.71
31	0.67
32	0.63
33	0.59
34	0.55
35	0.51

3.8 CAPACITANCE.- When measured as specified in 4.6.5, the capacitance shall be within the capacitance tolerance shown in the type designation.

3.9 DISSIPATION FACTOR.- When measured as specified in 4.6.6, the dissipation factor shall be not more than 1.0 percent.

3.10 VIBRATION.- As a result of the test specified in 4.6.7, there shall be no mechanical damage, and the electrical indication during the test shall show no evidence of intermittent contacts or open or short-circuiting.

3.11 TEMPERATURE AND IMMERSION CYCLING.- When tested as specified in 4.6.8, capacitors shall meet the following requirements:

Dielectric Withstanding Voltage

Terminal to terminal----------As specified in 3.5.

Insulation resistance

Terminal to terminal----------Not less than 4,000 megohms.

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3.12 HUMIDITY.- When tested as specified in 4.6.9, capacitors shall meet the following requirements:

Dielectric Withstanding Voltage

Terminal to terminal----------As specified in 3.5.

Insulation Resistance

Terminal to terminal----------Not less than 4,000 megohms.

3.13 LOW TEMPERATURE VOLTAGE TEST AND CAPACITANCE CHANGE WITH TEMPERATURE.- When tested as specified in 4.6.10, capacitors shall withstand the specified potential without any indication of breakdown or arcing during the test. As a result of the test, there shall be no open- or short-circuiting nor permanent damage to the capacitor. The capacitance change shall be as shown in detail specification.

3.14 LIFE.- When tested as specified in 4.6.11 capacitors shall meet the following requirements:

Insulation resistance or "megohms times microfarads" as applicable----------	Not less than 50 percent of the value specified in 3.7.1.
Capacitance----------	Change not more than 5 percent from the initial value obtained when measured as specified in 4.6.5.
Dissipation Factor----------	As specified in 3.9.

There shall be no mechanical failure nor open- or short-circuiting during life test or after short term vibration following life test.

3.15 SHOCK.- When tested as specified in 4.6.12, capacitors shall withstand the specified potential without momentary or intermittent arcing nor any other indication of breakdown or arcing during the test. As a result of the test, there shall be no open- or short-circuiting or mechanical damage to the capacitor.

3.16 LEAD BENDING.- As a result of the test specified in 4.6.13, capacitors shall show no mechanical damage.

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3.17 FLUX.- No flux of any kind shall be used in the assembly of the capacitors; however, exception is permitted for preliminary tinning or solder coating of the terminal lead wires. If acid or acid salts are used, as permitted above, they shall be completely neutralized and removed after use.

3.18 MARKING.- Each capacitor shall be marked in accordance with Standard MIL-STD-130, with the type designation, manufacturer's name or symbol, lot number, capacitance, and voltage rating. There shall be no space between the symbols which comprise the type designation. If space limitation requires it, the capacitance and voltage rating may be omitted and the type designation may appear on two or three lines. In this event, the type designation shall be divided between the style and the capacitance symbols, or between the style, capacitance and characteristic symbols as shown in the following examples:

CTM	CTM
104 VAK	104
	VAK

Marking shall remain legible after all tests.

3.19 WORKMANSHIP.- Capacitors shall be manufactured and processed in a careful and workmanlike manner in accordance with good design and sound practice and shall meet all requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 SUPPLIER'S RESPONSIBILITY.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. Inspection records of the examinations and tests shall be kept complete and available to the government as specified in the contract or order.

4.2 CLASSIFICATION OF TESTS.- The inspection and testing of capacitors shall be classified as follows:

a. Qualification testing	See 4.3
b. Acceptance tests	See 4.5

4.3 QUALIFICATION TESTING

4.3.1 SAMPLE.- The number of specimens comprising a sample of capacitors to be submitted for qualification tests shall be as specified in the appendix to this specification.

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4.3.2 TEST ROUTINE.- The specimens will be subjected to the qualification tests listed in table IV in the order shown. Two specimens in a sample shall be subjected to visual and mechanical examination (internal). The remaining specimens will be subjected to the subsequent examination and tests in group I. The specimens shall then be divided into three remaining groups as shown in table IV, and subjected to the tests for their particular group.

4.3.3 DEFECTIVES.- Defectives in excess of those allowed in table IV will be cause for refusal to grant qualification.

4.3.4 QUALIFICATION APPROVAL REVIEW.- The Qualified Products List shall be reviewed periodically by the qualifying agency for the possible removal of the manufacturers who are no longer producing to the specification or whose lot rejection rate during group B testing exceeds 15 per cent. A manufacturer shall be considered out of production if he fails to submit the required reports.

4.4 TEST CONDITIONS.- Unless otherwise specified herein, all tests shall be made at room ambient temperature, pressure, and humidity.

4.5 ACCEPTANCE TESTS.- Acceptance tests shall consist of groups A, B, and C.

4.5.1 CONTRACTOR'S TESTS.- The manufacturer shall subject each capacitor to the following tests:

a. Capacitance

b. Dielectric withstanding voltage

4.5.2 INSPECTION LOT.- An inspection lot shall be as specified in Standard MIL-STD-105, and shall consist of capacitors having the same voltage rating and number of dielectric layers, and having a maximum ratio of 20 to 1 between the maximum and minimum nominal capacitance values. Each lot shall be kept separate from every other lot. Test data shall be submitted by the manufacturer with each lot.

4.5.3 GROUP A. INSPECTION.- Group A inspection shall consist of the examination and tests specified in table V. Statistical sampling and inspection shall be in accordance with Standard MIL-STD-105. The acceptance quality levels (AQL) shall be as specified in table V.

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TABLE IV. - Qualification Inspection

Examination or test	Requirement Paragraph	Method Paragraph	No. of specimens to be inspected	No. of defectives allowed (See Note 1)
Group I				
Visual and Mechanical Inspection	3.2, 3.3 to 3.4.2.2 incl. and 3.17 to 3.19 incl.	4.6.1		
Marking, workmanship		4.6.1		
Dielectric withstanding voltage	3.5	4.6.2		
Insulation Resistance	3.7	4.6.4	102	0
Capacitance	3.8	4.6.5		
Dissipation Factor	3.9	4.6.6		
Flashover	3.6	4.6.3		
Group II				
Vibration	3.10	4.6.7	12	1
Temperature & immersion cycling	3.11	4.6.8		1
Group III				
Low Temperature Voltage Test and capacitance change with temperature.	3.13	4.6.10	12	1
Shock	3.15	4.6.12		
Group IV				
Lead Bend (see note 2)	3.16	4.6.13	6	
Humidity	3.12	4.6.9	75	1
Life	3.14	4.6.11.2		

Notes: 1 - A specimen having one or more defects will be considered as a single defective.

2 - In the event of a failure substitute a new unit.

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TABLE V. - Group A Inspection

Examination or Test	Requirement Paragraph	Method Paragraph	AQL (percent defective)
Subgroup 1			
Dielectric Withstanding Voltage.	3.5	4.6.2	0.40
Subgroup 2			
Insulation Resistance	3.7	4.6.4	
Capacitance	3.8	4.6.5	0.10
Dissipation Factor	3.9	4.6.6	
Subgroup 3			
Visual and mechanical examination Material	3.3 and 3.3.1		
Design and construction	3.4 to 3.4.2.2		
Physical Dimension	3.2		1.0
Marking	3.18		
Workmanship	3.19		

4.5.3.1 RESUBMITTED LOTS.- Rejected lots may be submitted for Government acceptance only if the manufacturer performs 100 percent testing of capacitors in the lot for those characteristics that were defective and resulted in rejection of the lot, removes all defective units, and resubmits the lot for acceptance testing. If dielectric with-standing voltage is one of the properties being rechecked, the lot shall also be tested for insulation resistance, capacitance, and dissipation factor. Resubmitted lots shall be kept separate from new lots, and shall be tested using the tightened inspection procedure of Standard MIL-STD-105.

4.5.4 GROUP B INSPECTION.- Group B inspection shall consist of the tests specified in table VI.

TABLE VI .- Group B Inspection

Test	Requirement Paragraph	Method Paragraph
Performance Check	3.14	4.6.11.1

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4.5.4.1 SAMPLING PROCEDURE.- The sampling procedure shall be as follows:

a. Each lot shall be sampled at the percentage stated in table VII.

b. The percentage shall be determined from the accumulated production at the beginning of the production lot.

c. Capacitance values selected shall represent the approximate ratio of the capacitance values in the lot. No sample from any lot shall be less than 30 specimens.

TABLE VII.- Sampling Schedule

Consecutively Produced Items	Percent of Production to be tested per lot. Percent
0-1000	30
1001-2000	20
2001-3000	17
3001-4000	14
4001-5000	12
5001-6000	10
6001-7000	8
7001-8000	7.5
8001-9000	6
9001- and over	5

4.5.4.2 QUALITY CONTROL REQUIREMENTS.- The manufacturer shall set up a process control system (see figure 1) and maintain records sufficiently detailed to meet all requirements of this specification.

4.5.4.2.1 PRODUCTION DAY.- The working days that the facility is closed shall be recorded as production days (see figure 1) and entered into the Production & Test Record as zero production and zero tested.

4.5.4.2.2 PRODUCTION AND TEST RECORD.- A production and test record shall be maintained similar to figure 1. The number of capacitors produced in each production lot shall be recorded. The number of capacitors selected for group B inspection testing shall be selected per 4.5.4.1 and recorded at the time testing starts. The method for calculating the numbers to be placed in the summation columns is shown and explained in figure 1.

4.5.4.2.3 TEST RECORD CHART.- A test record chart shall be maintained similar to figure 2. The summation of capacitors placed on test during the previous 30 production days shall be entered in the appropriate space along the bottom of the graph (see figure 2). Points for the control limit line shall be calculated or read from the graph on figure 3, and appropriately marked on the graph (see figure 2). The accumulated number of failure shall be taken from the Production and Test Record and plotted at the appropriate place (see figure 2). The day on which a lot (or lots) is (are) rejected shall be legibly marked (see figure 2).

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Production day 1/	Lot Number	Number per lot	Summation produced 2/	Sample Size	N = Summation of Capacitors tested 2/	Number failed	Summation of failures 2/	Control line 1 3/	Control line 2 3/
1	1	600	600	180	180	1	1	-----	-----
2	2	800	1400	240	420	2	3	8.06	1.71
3	3	600	2000	120	540	1	4	9.9*	2.80
4	4	750	2750	126	668	2	6	12.0	4.01
5	5	650	3400	111	779	1	7	13.75	5.10
6	6	600	4000	84	863	1	8	15.03	5.93
7	7	700	4700	84	947	1	9	16.35	6.78
8	8	300	5000	36	983	1	10	16.86	7.16
9	9	250		30		0			
9	10	750	6000	75	1088	1	11	18.46	8.25
10	11	1000	7000	80	1168	1	12	19.7	9.11
11	12	400		30		0			
11	13	600	8000	45	1243	0	12	20.84	9.91
12	14	1000	9000	60	1303	0	12	21.74	10.55
13	15	700		35		0			
13	16	800	10500	40	1378	0	12	22.84	11.33
14	17	900	11400	45	1423	5	17	23.53	11.84
15	18	600	12000	30	1453	0	17	23.98	12.15
16	19	600	12600	30	1483	0	17	24.43	12.47
17	20	600	13200	30	1513	0	17	24.87	12.79
18	21	600	13800	30	1543	0	17	25.33	13.11
19	22	900	14700	45	1588	0	17	25.97	13.63
20	23	1000	15700	50	1678	0	17	27.29	14.61
21	24	800		40		0			
21	25	700	17200	35	1713	0	17	27.82	14.99
30	34	800	24280	40	2062	1	26	33.04	18.90
31	35	1000	24680	50	1932	0	25	31.08	17.42
32	36	700	24580	35	1727	0	23	28.0	15.15

1/See 4.5.4.2.1

2/The previous 30 production days is the limit that is considered for the summation columns. The 1st day's numbers are subtracted from the 30th days summation numbers, then the 31st-day's numbers are added to arrive at the 31st-day's summations (see examples below).

EXAMPLES:

Summation Produced -

31 day (24,280-600+1000)= 24680

32 day (24,680-800+700) = 24580

N-Summation of Capacitors tested -

31 day (2,062-180+50) = 1932

32 day (1,932-240+35) = 1727

Summation of failures:

31 day (26 -1 + 0) = 25

32 day (25 -2 + 0) = 23

3/Values for the control lines may be read from figure 3 or may be calculated as follows:

Control line 1 = $\frac{N}{75} + 1.04\sqrt{\frac{N}{75}}$

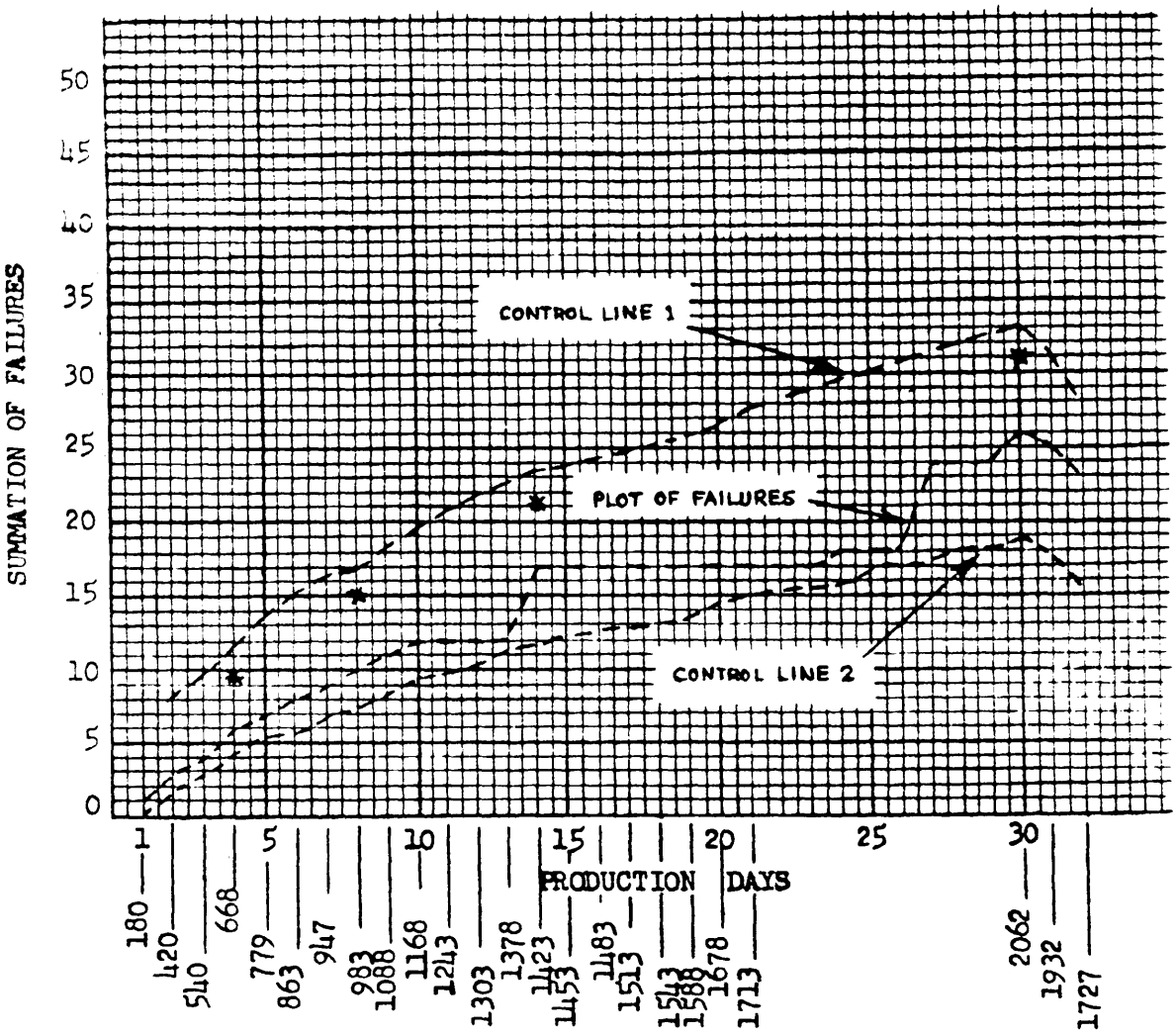
Control line 2 = $\frac{N}{75} - 1.64\sqrt{\frac{N}{75}}$

} See Footnote 2

The control limit lines have no meaning below N = 300. (see figure 3)

Figure 1. **Production and test record.**

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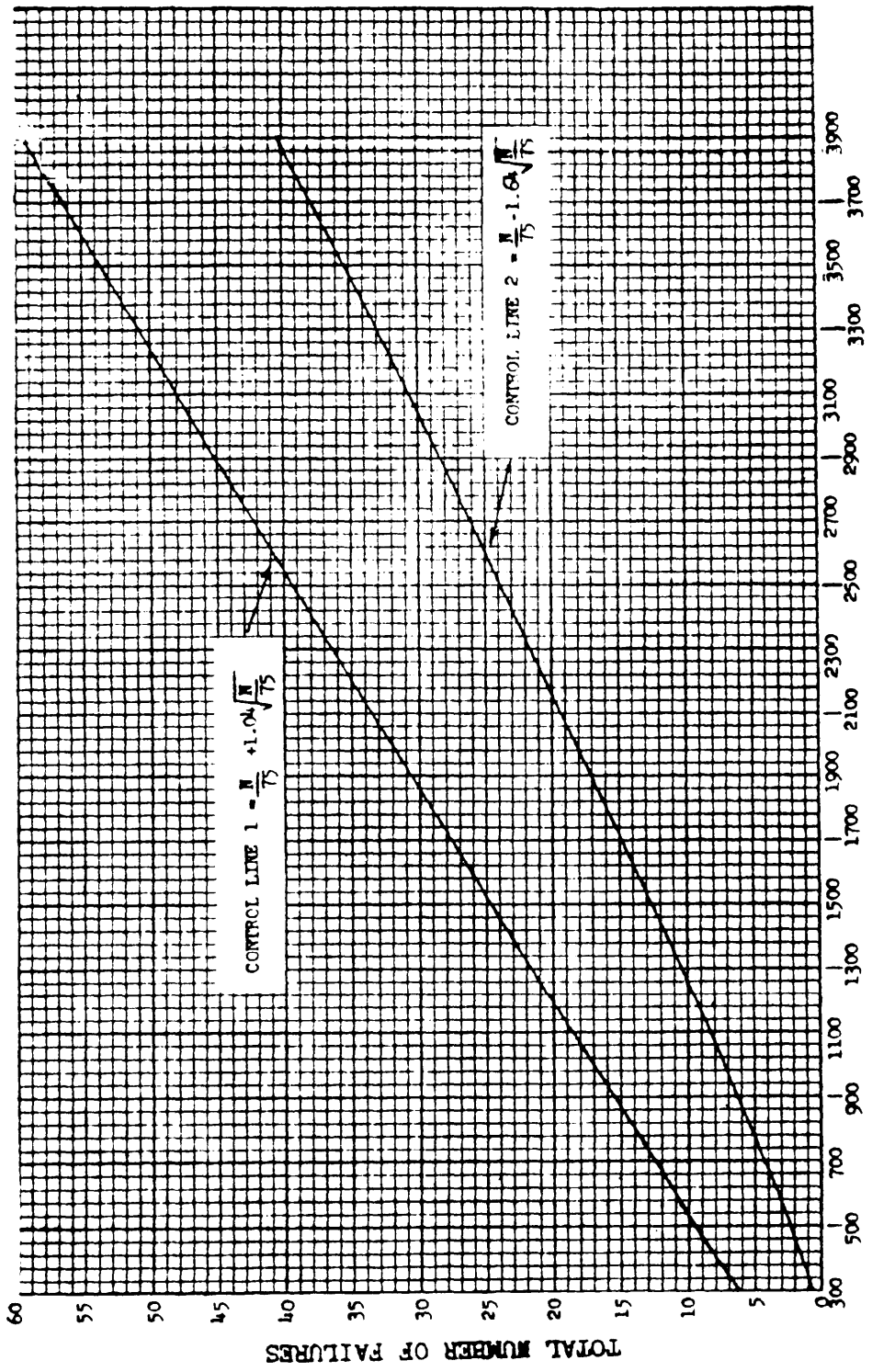
N= SUMMATION OF CAPACITORS TESTED IN PREVIOUS 30 PRODUCTION DAYS

* = Production day on which a lot(s) was (were) rejected.

Note - The individual points of the control limit lines and the plot of failures may be connected for ease of reading.

FIGURE 2. Suggested Test Record Chart

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N (Number of Capacitors Tested)

Figure 3 shall be used for establishing points for the test record chart. (see figure 2)

FIGURE 3 CONTROL LIMITS

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4.5.4.3 ALLOWABLE FAILURE.- Allowable failures shall be 1 failure per each 75 specimens tested of each lot. For a sample lot of less than 75 units, the allowable failures shall be zero. The manufacturer, at his own option, may add specimens to any test sampling in order to complete a group evenly divisible by 75. The decision involving the use of the option shall be made prior to starting the test. Any lot exceeding the allowable failure rate shall not be resubmitted for Government acceptance.

4.5.4.3.1 ACCUMULATED FAILURES.- When it is determined that the accumulated failure plot lies above the control line No. 1 (see figure 3), the next production lot manufactured shall be sampled at the rate of 30 percent. The lots previously produced, but as yet undetermined as to acceptance or rejection, shall be accepted or rejected depending upon the results of group B testing for the individual lots in question. A new "Production and Test Record" and a "Test Record Chart" similar to figures 1 and 2 shall be initiated.

4.5.4.4 REPORTED INFORMATION.- The manufacturer will submit reports to the qualifying agency as follows:

a. Every 30 days.- A reproduction of the "Production and Test Record" (see 4.5.4.2.2) and the "Test Record Chart" (see 4.5.4.2.3) for the previous 30 days.

b. When the accumulated failure plot lies above control line No. 1 on the Test Record Chart (see figure 2) the manufacturer will report the following:

1. The voltage and capacitance rating for each capacitor.

2. Time after start of test that failures occurred, if known.

3. Reason for failure, if known.

4. Explanation or remedial action proposed by the manufacturer.

4.5.5 GROUP C INSPECTION.- Group C inspection shall consist of the tests specified in table VIII in the order shown.

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TABLE VIII.- Group C Inspection

TEST	Requirement Paragraph	Method Paragraph	Number of Specimens	Allowable Failures
SUBGROUP No. 1 every two weeks				
Flashover	3.6	4.6.3	48	2
Vibration	3.10	4.6.7		
Temp. & immersion cycling	3.11	4.6.8		
SUBGROUP No. 2 every two weeks				
Low temp. voltage test & capacitance chg. with Temp.	3.13	4.6.10	48	2
Shock	3.15	4.6.12		
SUBGROUP No. 3 every three months				
Lead Bend	3.16	4.6.13	48	2
Humidity	3.12	4.6.9		
Life	3.14	4.6.11.2		

4.5.5.1 SAMPLING PROCEDURE.- Forty-eight specimens shall be taken from production every 2 weeks for each of subgroups one and two and every 3 months for subgroup three as listed in table VIII and subjected to the tests in the order shown. The maximum and minimum case size manufactured during that two weeks shall be represented in the sample in at least the approximate ratio of production. Allowable failure rate shall be as shown in table VIII.

4.5.5.2 DISPOSITION OF SAMPLE UNITS.- Sample units subjected to group C inspection shall not be delivered on the contract or order.

4.5.5.3 NONCOMPLIANCE.- If a sample fails to pass group C inspection, the manufacturer shall take corrective action on the process and all units of the product which can be corrected and which were manufactured with the same conditions, materials, processes, etc. and are considered subject to the same test failure. Acceptance testing shall be discontinued until corrective action has been taken. After the corrective action, the sample unit shall be subjected to the necessary group C inspection. (All inspections, or the failed inspections, at the option of the Government.) Groups A and B inspection may be reinstituted; however, final acceptance shall be withheld until the group C inspection has shown that the corrective action was successful.

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4.6 TEST METHODS

4.6.1 VISUAL AND MECHANICAL EXAMINATION.- Capacitors shall be inspected to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements.

4.6.2 DIELECTRIC WITHSTANDING VOLTAGE.- The dielectric withstanding voltage test shall be performed in accordance with method 301 of Standard MIL-STD-202. Refer to the applicable detail specification for the test voltage level.

4.6.3 FLASHOVER.- Capacitors shall be tested in accordance with method 105 of Standard MIL-STD-202, except the following shall apply:

a. Method of Mounting - By normal mounting means.

b. Test Condition - The pressure shall be equivalent to an altitude of 80,000 feet (0.52 inches Hg.)

c. Test During Subjection to Reduced Pressure - A potential 125 percent of rated voltage shall be applied for at least 1 minute between terminals. A suitable means shall be used to detect momentary or permanent breakdown.

4.6.4 INSULATION RESISTANCE.- Capacitors shall be tested in accordance with method 302 of Standard MIL-STD-202. The following details and exceptions shall apply:

a. Test Potential - Rated voltage.

b. Terminal to Terminal - Insulation resistance shall be measured between terminals at a temperature of 25° ± 3°C or corrected thereto per table III, and at 125° ± 3°C.

4.6.5 CAPACITANCE.- Capacitors shall be tested in accordance with method 305 of Standard MIL-STD-202. The following details shall apply:

a. Test Frequency - 1,000 ± 100 cycles per second (cps).

b. Measurement Accuracy - Shall be within ± 1 percent.

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4.6.6 DISSIPATION FACTOR.- The dissipation factor of each capacitor shall be measured at an ac voltage not greater than 20 percent of the rated dc voltage, at a frequency of 1,000 ± 100 cps. Measurement accuracy shall be within ± 2 percent.

4.6.7 VIBRATION.- The capacitor shall be rigidly mounted by the entire body length to the vibration test apparatus. The capacitors shall be tested in accordance with method 204 of Standard MIL-STD-202, condition B. The vibration shall be for 4 hours in each of two directions, parallel and perpendicular to the major axis. During the last cycle in each direction, an electrical measurement shall be made to determine intermittent contacts or open-or short-circuiting.

4.6.8 TEMPERATURE AND IMMERSION CYCLING

4.6.8.1 TEMPERATURE CYCLING.- Capacitors shall be tested in accordance with method 102 of Standard MIL-STD-202. The following details and exceptions shall apply:

a. Test-condition letter - C, except that during step 1, capacitors shall be conditioned at $-55^{\circ}\ ^{+0^{\circ}}_{-3^{\circ}}$C.

b. Measurements before and after cycling - Not applicable.

4.6.8.2 IMMERSION CYCLING.- Within 4 to 24 hours after completion of temperature cycling, capacitors shall be tested in accordance with method 104 of Standard MIL-STD-202. The following details and exceptions shall apply:

a. Test-condition letter - B.

b. Measurements after final cycle. In addition, dielectric strength and insulation resistance at 25°C of all capacitors shall be measured as specified in 4.6.2 and 4.6.4 respectively.

After this test, capacitors shall be visually examined for damage to case and obliteration of marking.

4.6.9 HUMIDITY.- Capacitors shall be tested in accordance with method 102, test condition A, of Standard MIL-STD-202, except that no measurements shall be made before and after cycling. Capacitors shall then be tested in accordance with method 103, test condition A of Standard MIL-STD-202. The following details and exceptions shall apply:

a. Temperature of the chamber shall be maintained at 65° ± 2°C.

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b. Measurements after exposure - Upon completion of the exposure period, the capacitors shall be conditioned at 25° ± 5°C and a relative humidity of 50 ± 5 percent for a period of at least 22 hours but not more than 24 hours. Dielectric strength and insulation resistance at 25°C of all capacitors shall be measured as specified in 4.6.2 and 4.6.4 respectively.

After the test, capacitors shall be visually examined for damage to case and obliteration of marking.

4.6.10 LOW TEMPERATURE VOLTAGE TEST AND CAPACITANCE CHANGE WITH TEMPERATURE.

4.6.10.1 LOW TEMPERATURE VOLTAGE TEST.- Capacitors shall be placed in a chamber maintained at -55° ± 3°C, and rated dc voltage shall be applied at this condition for 48 ± 4 hours. The air within the conditioning chamber shall be circulated.

4.6.10.2 CAPACITANCE CHANGE WITH TEMPERATURE.- At the conclusion of the test specified in 4.6.10.1, capacitance measurements shall be made as specified in 4.6.5, except that measurements shall be at -55° ± 3°C, 25° ± 5°C, and 125° ± 3°C. The -55°C measurement shall be made before the capacitors are removed from the conditioning chamber. The measurements at each temperature shall be recorded when two successive readings taken at 5-minute intervals indicate no change in capacitance.

4.6.11 INTEGRATED TIME-VOLTAGE TESTS

4.6.11.1 PERFORMANCE CHECK.- Capacitors shall be subjected to 140 percent of the rated dc voltage at the high ambient test temperature for 250 ± 8 hours. During the conditioning, capacitors shall be separated by a distance of not less than 1 inch. Adequate circulation of air shall be provided to prevent the temperature within 6 inches of any capacitor from departing more than + 2°C from the nominal ambient temperature of the chamber. Radiation shall not be used as a means of heating the chamber. The surge current shall not exceed 1 ampere. When necessary, a suitable current-limiting resistor shall be inserted into the circuit. At the conclusion of this test, the capacitors shall be returned to the test conditions specified in 4.4. Insulation resistance at 25°C, capacitance, and dissipation factor shall then be as specified in 4.6.4, 4.6.5, 4.6.6, respectively. After measurements specified above, the capacitors shall be rigidly mounted by their normal means on a vibration test apparatus. The capacitors shall be tested in accordance with method 201 of MIL-STD-202, condition A. The vibration shall be 45 minutes in each of two directions,

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parallel and perpendicular to the major axis. During the last 3 minutes of vibration in each direction, an electrical measurement shall be made to determine intermittent contacts or open-or short-circuiting.

4.6.11.2 LIFE.- Capacitors shall be subjected to the rated d-c voltage, at the ambient test temperature of 125°C, for 2,000 ± 24 hours. During the conditioning, capacitors shall be separated by a distance of not less than 1 inch. Adequate circulation of air shall be provided to prevent the temperature within 6 inches of any capacitor from departing more than 2°C from the nominal ambient temperature of the chamber. The surge current shall be limited to 1 ampere, maximum. When necessary, a suitable current limiting resistor shall be inserted into the circuit. At the conclusion of this test, the capacitor shall be returned to the test conditions specified in 4.4. Insulation resistance at 25°C, capacitance and dissipation factor shall then be measured as specified in 4.6.4, 4.6.5 and 4.6.6, respectively.

4.6.12 SHOCK.- The capacitors shall be securely mounted by the entire body length to a test fixture. The test fixture shall then be securely mounted to the carriage of a shock machine designed and fabricated in accordance with Specification MIL-S-4456. The capacitors shall then be subjected to 18 impact shocks of the intensity of 50g's. Each shock impulse shall have a time duration of 11 ± 1 milliseconds. The shocks shall be applied in the following axes:

a. Vertically, 3 shocks in each direction.

b. Parallel to the major horizontal axis, 3 shocks in each direction.

c. Parallel to the minor horizontal axis, 3 shocks in each direction.

Electrical integrity shall be monitored during shock. A cathoderay oscilloscope, or other comparable means shall be used as the indicating device.

4.6.13 LEAD BEND.- Wire-lead terminals shall be bent through 90° at a point 1/4 inch from the body of the capacitor, with the radius of curvature at the bend approximately 1/32 inch. The terminals shall be clamped to within 3/64 ± 1/64 inch of the bend. The body of the capacitor or the clamped terminal shall then be rotated about the original axis of the bent terminal through 360° in alternating directions for three rotations at the rate of approximately five seconds per rotation.

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5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND PACKAGING

5.1.1 LEVEL A.- Capacitors shall be individually protected and unit-packaged in accordance with method III of Specification MIL-P-116. Capacitors shall be cushioned and packaged in 50-pound kraft paper envelopes, sealed. Unless otherwise specified (see 6.2), five unit packages or a multiple thereof shall be further packaged in intermediate containers conforming to Specification PPP-B-566, PPP-B-676, or PPP-B-665. The gross weight of the intermediate container shall not exceed 10 pounds.

5.1.2 LEVEL C.- Capacitors shall be afforded preservation and packaging in accordance with the manufacturer's replacement-parts practice.

5.2 PACKING

5.2.1 LEVEL A.- Capacitors packaged as specified (see 6.2) shall be packed in overseas-type wirebound wood, wood-cleated fiberboard, wood-cleated plywood, nailed wood, fiber (class 2 or 3, as specified (see 6.2)), or wood-cleated paper-overlaid boxes conforming to Specifications PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, and MIL-B-10377, respectively, at the option of the manufacturer. Shipping containers shall have case liners conforming to Specification MIL-L-10547; the case liners shall be closed and sealed in accordance with the appendix thereto. Case liners for boxes conforming to Specification PPP-B-636 may be omitted provided the center and edge seams and manufacturers' joints are sealed with tape, at least 1-1/2 inches wide, conforming to type III, class 1, of Specification PPP-T-60. Box closures and strapping shall be as specified in the applicable box specification or appendix thereto. Fiber boxes conforming to Specification PPP-B-636 may be banded with tape conforming to type IV of Specification PPP-T-97 and appendix thereto in lieu of steel straps. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 LEVEL B.- Capacitors packaged as specified (see 6.2) shall be packed in domestic-type wirebound wood, wood-cleated fiberboard, wood-cleated plywood, nailed wood, fiber, or wood-cleated paper-overlaid boxes conforming to Specifications PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, and MIL-B-10377, respectively, at the option of the manufacturer. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

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5.2.3 LEVEL C.- Capacitors packaged as specified shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Consolidated Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.3 MARKING.- In addition to any special marking required by the contractor or order, unit packages, intermediate packages, and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 INTENDED USE.- The capacitors covered by this specification are intended for use in any equipment where higher orders of reliability are required and are primarily for filter, bypass, or blocking purposes where the a-c component of the impressed voltage is small with respect to the d-c voltage rating.

6.1.1 This capacitor adapts itself to encapsulation techniques for applications where extreme humidity conditions are encountered.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Complete type designation (see 1.2.1. and 3.2).

c. Levels of preservation, packaging, and packing and applicable marking (see section 5).

d. Number of unit packages if other than specified in 5.1.1.

6.3 QUALIFICATION.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is Commander, Wright Air Development Division, Attn: WWDPVE-2, Wright-Patterson Air Force Base, Ohio, and information pertaining to qualification of products may be obtained from that activity.

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6.4 MOUNTING.- Capacitors covered by this specification should be mounted by a bracket or clamp when vibration requirements exceed 55 cycles or when high shock is likely to be encountered in service.

6.4.1 Capacitors covered by this specification should be mounted in such a manner that there is no torsional stress on the lead wires.

6.5 APPLICATION INFORMATION.- Capacitors procured under the quality assurance afforded by this specification will experience failures at a rate depending almost exclusively upon the manner in which the capacitors are used. The life of the capacitor is primaril dependent upon temperature and voltage. Capacitors should not be used above the rated voltage and rated temperature where reliability is a factor.

6.5.1 DC CAPACITORS.- The energy content of a dc capacitor when fully charged is determined by the following formula:

$$W = \frac{CE^2}{2}$$

Where

W = energy content in watt-seconds.

C = nominal capacitance in farads.

E = dc voltage rating in volts at 40°C.

6.6 DC VOLTAGE.- A dc voltage is unidirectional voltage in which the changes in value are either zero or so small that they may be neglected.

6.7 AC COMPONENT.- The rating given is the steady state dc voltage, or the sum of the dc voltage and the peak ac voltage, provided that the peak ac voltage does not exceed 20 percent of the rating at 60 cps, 15 percent at 120 cps, or 1 percent at 10,000 cps. Where heavy transient or pulse currents are encountered, the requirements of this specification are not sufficient to guarantee satisfactory performance and due allowance must therefore be made in the selection of a capacitor.

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APPENDIX

PROCEDURE FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 This appendix details the procedure for submission of samples, with related data, for qualification testing of capacitors covered by this specification.

20. SUBMISSION

20.1 SAMPLE

20.2 TEST DATA.- Each submission shall be accompanied by test data covering the nondestructive tests listed in table IV which have been performed on the submitted sample units. The performance of the destructive tests by the supplier on a duplicate set of sample units is encouraged, although not required. All test data shall be submitted in duplicate.

20.3 DESCRIPTION OF ITEMS.- The supplier shall submit a detail description of the materials and constructional features of the capacitors being submitted for qualification, including information as to whether they are impregnated or filled, the type and quantity of the impregnant or filling compound; the type, thickness, and number of layers of plastic film and foil; the material, thickness, and applied finish of the case, as well as details on the end seal and terminal assembly.

30. EXTENT OF QUALIFICATION

30.1 Each submission will confer eligibility for qualification for the combination of style and circuit diagram represented, except that any capacitance tolerance may also represent all other capacitance tolerances listed in the specification. Voltage-rating qualification and watt-second-rating qualification will be restricted to those ratings equal to and less than the voltage rating and watt-second rating represented in the submission.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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